

LINUX COMMAND CHEAT SHEET

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1. LINUX DIRECTORY STRUCTURE

/ - Root directory of the system
/home - User home directories
/etc - System configuration files
/var - Variable files (logs, spool, cache)
/usr - User installed programs & data
/tmp - Temporary files
/opt - Optional add-on applications
/root - Home directory for root user

/
├── home
├── etc
├── var
├── usr
├── tmp
├── opt
└── root

2. NAVIGATION COMMANDS

Command	Description	Example
pwd	Print working directory	pwd
ls	List files and directories	ls
ls -l	Long listing format	ls -l
ls -a	List all files (including hidden)	ls -a
cd <dir>	Change directory	cd /etc
cd ..	Go one directory back	cd ..
cd ~	Go to home directory	cd ~
cd -	Go to previous directory	cd -
clear	Clear terminal screen	clear
history	Show command history	history

3. FILE & DIRECTORY COMMANDS

Command	Description	Example
touch <file>	Create an empty file	touch a.txt
mkdir <dir>	Create a directory	mkdir mydir
mkdir -p <dir>	Create nested directories	mkdir -p a/b/c
rmdir <dir>	Remove empty directory	rmdir emptydir
rm <file>	Remove a file	rm file.txt
rm -r <dir>	Remove directory (recursively)	rm -r mydir
rm -rf <dir>	Force remove directory	rm -rf mydir
cp <src> <dest>	Copy file	cp a.txt b.txt
cp -r <src> <dest>	Copy directory recursively	cp -r mydir bakdir
mv <src> <dest>	Move / Rename file or directory	mv old.txt new.txt
tree	Show directory tree (if installed)	tree

4. FILE VIEWING COMMANDS

Command	Description	Example
cat <file>	Display file content	cat file.txt
less <file>	View file page by page	less file.txt
more <file>	View file page by page (older)	more file.txt
head <file>	Show first 10 lines	head file.txt
head -n <num> <file>	Show first n lines	head -n 20 file.txt
tail <file>	Show last 10 lines	tail file.txt
tail -n <num> <file>	Show last n lines	tail -n 20 file.txt
tail -f <file>	Follow / watch file (live logs)	tail -f app.log
nl <file>	Number all lines	nl file.txt

5. SEARCH COMMANDS

Command	Description	Example
find <path> -name <name>	Search files in a directory	find /home -name test.txt
locate <name>	Search file using database	locate test.txt
which <cmd>	Show path of a command	which python
whereis <cmd>	Show binary, source, man info	whereis java
grep <pattern> <file>	Search pattern in a file	grep "error" app.log
grep -r <pattern> <dir>	Search pattern in directory	grep -r "fail" /var/log

grep Example :

grep "error" app.log → Search the word "error" in app.log

6. PERMISSIONS

chmod <permissions> <file/dir>

r (read) = 4

w (write) = 2

x (execute) = 1

Examples :

chmod 755 script.sh → rwxr-xr-x

chmod 644 file.txt → rw-r--r--

chmod 777 file.txt → rwxrwxrwx

Permission Table

Type	r	w	x	Value
Read	Yes	-	-	4
Write	-	Yes	-	2
Execute	-	-	Yes	1

Permission Representation

Owner	Group	Others
rwx	r-x	r-x
7	5	5

7. OWNERSHIP

Command	Description	Example
chown <user> <file/dir>	Change owner	chown aman file.txt
chgrp <group> <file/dir>	Change group	chgrp developers file.txt
chown <user>:<group> <file/dir>	Change owner and group	chown aman:dev file.txt

8. FILE COMPRESSION

Command	Description	Example
zip <file.zip> <files>	Create zip file	zip files.zip file.txt
unzip <file.zip>	Extract zip file	unzip files.zip
tar -cvf <file.tar> <files>	Create tar file	tar -cvf data.tar data/
tar -xvf <file.tar>	Extract tar file	tar -xvf data.tar
tar -czvf <file.tar.gz> <dir>	Create tar.gz (compressed)	tar -czvf logs.tar.gz logs/
tar -xzvf <file.tar.gz>	Extract tar.gz file	tar -xzvf logs.tar.gz

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9. PROCESS MANAGEMENT

Command	Description	Example
ps	Show current processes	ps
ps -ef	Show all processes (full format)	ps -ef
top	Display processes in real time	top
htop	Interactive process viewer (if installed)	htop
kill <pid>	Terminate process by PID	kill 1234
kill -9 <pid>	Force kill process	kill -9 1234
pkill <name>	Kill process by name	pkill java
jobs	Show background jobs	jobs
bg %<job_id>	Resume job in background	bg %1
fg %<job_id>	Bring job to foreground	fg %1

Process Flow :



10. DISK USAGE & SYSTEM INFO

Command	Description	Example
df -h	Disk space usage (human readable)	df -h
du -sh <dir>	Directory size summary	du -sh /var/log
du -ah --max-depth=1	Show size of subdirectories	du -ah --max-depth=1
free -h	Memory usage	free -h
uname -a	System information	uname -a
uptime	System uptime and load average	uptime
who -r	Show system run level	who -r

11. NETWORKING COMMANDS

Command	Description	Example
ping <host>	Check connectivity	ping google.com
curl <url>	Transfer data from URL	curl google.com
wget <url>	Download file from URL	wget file.zip
ssh <user>@<host>	Secure remote login	ssh aman@192.168.1.10
scp <src> <dest>	Secure copy (remote files)	scp file.txt user@host:/home
netstat -tulnp	Show open ports & connections	netstat -tulnp
ss -tulnp	Another way to see sockets	ss -tulnp
hostname	Show or set hostname	hostname
ip addr	Show IP addresses	ip addr
ifconfig	Network interface config	ifconfig

12. USER MANAGEMENT

Command	Description	Example
whoami	Show current logged in user	whoami
who	Show logged in users	who
id	Show user and group id	id
adduser <name>	Add a new user (Ubuntu)	sudo adduser aman
userdel <name>	Delete user	sudo userdel aman
passwd <user>	Change user password	passwd aman
sudo <cmd>	Run command as superuser	sudo apt update
su - <user>	Switch to another user	su - aman

13. ENVIRONMENT VARIABLES

Command	Description	Example
echo \$VAR	Display value of variable	echo \$PATH
export VAR=value	Set environment variable	export JAVA_HOME=/usr/lib/jvm
env	Show all environment variables	env
printenv	Another way to show env vars	printenv
unset VAR	Remove environment variable	unset VAR
source file.sh	Load variables from a file	source ~/.bashrc

PATH Example :

\$PATH → /usr/local/bin → /usr/bin → /bin → ...

System searches these directories (left to right) for a command.

14. PACKAGE MANAGERS

Ubuntu / Debian (APT)	RHEL / CentOS / Fedora (YUM/DNF)
• sudo apt update # Update package list • sudo apt upgrade # Upgrade packages • sudo apt install <pkg> # Install package • sudo apt remove <pkg> # Remove package • sudo apt autoremove # Remove unused pkgs • apt search <pkg> # Search package	• sudo yum update • sudo yum install <pkg> • sudo yum remove <pkg> • sudo dnf install <pkg> • sudo dnf remove <pkg> • yum search <pkg>

15. USEFUL SHORTCUTS

Shortcut	Description
Ctrl + C	Terminate current command
Ctrl + Z	Suspend current command (to background)
Ctrl + D	Logout / Exit / End of input
Ctrl + R	Search command history
Ctrl + L	Clear terminal screen
Tab	Auto-complete command or file name
!!	Run last command
!n	Run nth command from history

History is your
best friend!



Use ↑ and ↓
arrows to navigate
through history.

16. DEVELOPER / TEXT PROCESSING COMMANDS

Command	Description	Example
echo	Print text	echo "Hello World"
tee <file>	Write output to file & console	echo data tee out.txt
xargs	Build and execute command lines	cat file.txt xargs rm
sort <file>	Sort lines of text	sort names.txt
unig <file>	Remove duplicate lines	sort file.txt unig
wc <file>	Count lines, words, chars	wc file.txt
cut -d',' -f1 <file>	Cut columns from file	cut -d',' -f1 data.csv
awk '{print \$1}' <file>	Pattern scanning & processing	awk '{print \$1}' file.txt
sed 's/a/b/g' <file>	Stream editor (replace text)	sed 's/old/new/g' file.txt
diff file1 file2	Compare two files	diff file1 file2

17. QUICK INTERVIEW TIPS

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|-----------------------------|------------------------------------|
| ✓ cat vs less vs more | ✓ tar vs zip vs gzip |
| ✓ cp vs mv | ✓ df vs du |
| ✓ find vs locate | ✓ sudo vs su |
| ✓ kill vs kill -9 | ✓ apt vs yum/dnf |
| ✓ ps vs top vs htop | ✓ grep vs awk vs sed |
| ✓ chmod 755 vs 644 | ✓ Network troubleshooting commands |
| ✓ Absolute vs Relative Path | ✓ Permission calculation (rwx) |
| ✓ Soft Link vs Hard Link | |